

# AVISHKAR MAHESH PAWAR

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## Professional Summary

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Computer Science student with strong background in AI/ML infrastructure and high-performance computing. Built vector database implementing HNSW algorithm for semantic search on 1M+ vectors with 95%+ recall. Experience integrating OpenAI embeddings and LangChain for RAG pipelines. Proficient in Python, FastAPI, and optimizing computations using Numba JIT compilation. Published research paper on Sign Language Recognition using Machine Learning. Strong foundation in algorithms and data structures.

## Education

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**MIT Academy of Engineering, Pune**

2022 - 2026

*B.Tech. - Computer Engineering — CGPA: 9.24/10*

## Projects

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### AI-Powered Vector Database

*Python 3.11, FastAPI, NumPy, Numba*

- Built a vector database implementing HNSW algorithm for approximate nearest neighbor search on 1,000,000+ vectors with 95%+ recall and <100ms query latency
- Optimized distance computations using Numba JIT compilation, achieving 10x speedup in cosine similarity calculations for high-dimensional vectors
- Integrated OpenAI embeddings (text-embedding-ada-002) and LangChain to power RAG pipeline for AI applications with context-aware responses

### Event Streaming Platform

*Java 17, Spring Boot, Netty, Apache Kafka Protocol*

- Developed REST API using Spring Boot for cluster management, topic creation, and real-time monitoring with Prometheus integration
- Optimized producer batching and compression (Snappy/LZ4) reducing network bandwidth by 70%
- Containerized entire platform using Docker with multi-node deployment configuration

### MiniDB — Distributed Database with ACID Transactions

*C++17, Boost.Asio, Protocol Buffers, CMake*

- Developed production-grade storage engine with B+ Tree indexing supporting range queries and point lookups with  $O(\log n)$  complexity
- Implemented full query processing pipeline with SQL-like parser and executor supporting SELECT, INSERT, UPDATE, DELETE with JOIN operations across distributed tables
- Built MVCC (Multi-Version Concurrency Control) for snapshot isolation, enabling 1000+ concurrent transactions without read locks, with cost-based query optimization

## Technical Skills

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**Languages:** C++, Java, Python, JavaScript, TypeScript

**Frameworks:** Spring Boot, FastAPI, React, Node.js, Express.js

**AI/ML:** TensorFlow, PyTorch, OpenAI API, LangChain, NumPy, Numba

**Databases:** PostgreSQL, MySQL, MongoDB, Redis, Custom Database Implementations

**Tools:** Git, Docker, Linux, AWS, CMake, Maven, Prometheus

**Distributed Systems:** Kafka, gRPC, Protocol Buffers, Raft Consensus, Consistent Hashing

**CS Fundamentals:** Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Distributed Systems, System Design

## Experience

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### OSMOS

Jul 2025 - Present

*Software Developer*

- Worked on backend services using Node.js and developed applications using Python
- Contributed to product development from scratch, including designing, implementing, and testing core features

### Core-Decimal Solutions

Feb 2025 - Aug 2025

*Software Developer*

- Contributed to full-stack project using Vue.js, Express.js, and Sequelize with MVC architecture
- Developed reusable Vue.js components and integrated RESTful APIs with Axios

## Achievements & Certifications

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**GATE 2026:** Secured AIR 3124 with GATE Score 601 and 51.45/100 marks in CS — 3x rank improvement from GATE 2025 (AIR 9860)

**GATE 2025:** Qualified with AIR 9860, GATE Score: 453, Marks: 39.32/100

**First Place in Technodium 25:** Won first place in technical competition

**Certifications:** CCNA: Introduction to Networks, CCNA: Switching, Routing, and Wireless Essentials, CCNA: Enterprise Networking, Security, and Automation

## Extra-Curricular Activities

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**Outdoor Sports & Games:** Active participant in outdoor sports and recreational activities